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Layer 2 Bridge

Do you have devices that can't run ZeroTier that you want to access remotely? You can use a small Linux PC as a bridge between ZeroTier and physical networks.

> NOTE
> This topic is related to but different from using ZeroTier as a Layer 5 Service Proxy.

Assumptions

- You're doing this on your home network and can log in to your router and find the DHCP settings.
- You have a keyboard, monitor, and ethernet cable plugged into your Pi. Chances are high we'll break networking and lose access to the Pi.
- You're somewhat familiar with the command line and ssh.
- We're going to use systemd networking for this. You could probably adapt the concepts to a different Linux network configuration system if you have opinions about systemd.
- We used a Raspberry Pi 2 while writing this, but a Pi 3 or 4 should work fine. Anything running a Debian 10+ based distro should be fine. It doesn't have to be a Raspberry Pi, but some of these instructions might be Raspbian specific.

What you'll need

- Physical LAN Subnet
- Physical LAN DHCP Range
- ZeroTier Auto-Assign Range
- Default Gateway IP Address (the router)
- Bridge IP Address (will be statically assigned)
- Create a new ZeroTier network and get the ID. Keep an old network around for secondary way to connect any devices already using ZeroTier.
- The DHCP range and ZeroTier Auto-Assign range should be in the same subnet, but not overlap. You'd probably base this off what is already configured on your router.

An example plan

Name	Value	Referred to below as
Physical LAN Subnet	192.168.192.0/24	
Physical LAN DHCP RANGE	192.168.192.65 through 192.168.192.126	
ZeroTier Auto-Assign Range	192.168.192.129 through 192.168.192.190	$ZT_POOL
ZeroTier Managed Route	192.168.192.0/23	$ZT_ROUTE
Default Gateway IP Address	192.168.192.1	$GW_ADDR
Bridge IP Address	192.168.192.2/24 (or use DHCP)	$BR_ADDR
ZeroTier Network ID	d5e04297a1f0bbd70	$NETWORK_ID
ZeroTier Network Interface Name	zt3jnwghuq	$ZT_IF

Get your bridge device up and running

Follow the Raspberry Pi install instructions

- Install Raspbian OS
- Enable SSH

SSH into the Pi

It's easier to login via ssh now and copy/paste commands from the comfort of your own PC.

The DNS name might just work for you:

`ssh pi@raspberrypi.local` Or `ssh pi@<ip-address-of-pi>`

Update the Operating System

```
sudo apt update && sudo apt -y full-upgrade && sudo reboot
```

Log back in after it's done

Install ZeroTier

```
https://www.ZeroTier.com/download/

curl -s https://install.zerotier.com | sudo bash
```

Let's set some shell variables now

```
NETWORK_ID=<your-network-id>
BR_IF="br0"
BR_ADDR=<your-bridge-address>
GW_ADDR=<your-gateway-address>
```

Join ZeroTier Network

```
sudo zerotier-cli join $NETWORK_ID
```

We don't want ZeroTier to manage addresses or routes on `$ZT_IF`. We're doing it statically below, on the bridge interface.

```
sudo zerotier-cli set $NETWORK_ID allowManaged=0
```

Set one more variable

```
ZT_IF=<your-zt-interface-name>
```

Copy the `dev` name from the `listnetworks` output for `$ZT_IF`. It will be something like: zt3jvirser

Configure the device in ZeroTier Central

- Go to the Members section of the Network
- Open the Wrench icon for advanced settings and check
- Check Allow Bridging
- Check Do Not Auto Assign
- Authorize the member

Switch to systemd networking

> NOTE
> With Raspbian/Debian 12.0 Bookworm or later, `systemd-resolved` is not included and must be installed manually. https://www.debian.org/releases/bookworm/arm64/release-notes/ch-information.en.html#systemd-resolved
>
> Ensure this package is installed before removing `openresolv`, otherwise DNS resolution will likely be broken.

```
sudo apt install systemd-resolved
```

Remove existing network stuff

```
sudo apt remove --purge --auto-remove dhcpcd5 fake-hwclock ifupdown isc-dhcp
```

Enable systemd-networkd

```
sudo ln -sf /run/systemd/resolve/resolv.conf /etc/resolv.conf;
sudo systemctl enable systemd-networkd;
sudo systemctl enable systemd-resolved;
sudo systemctl enable systemd-timesyncd;
```

Configure interfaces

```
sudo zerotier-cli set $NETWORK_ID allowManaged=0
```

Write Network Configuration files. Puts ethernet and zerotier into the bridge and configures the bridge with a static IP. See below for DHCP configuration on the bridge.

```
cat << EOF | sudo tee /etc/systemd/network/25-bridge-br0.network
[Match]
Name=$BR_IF

[Network]
Address=$BR_ADDR
Gateway=$GW_ADDR
DNS=1.1.1.1
EOF

cat << EOF | sudo tee /etc/systemd/network/br0.netdev
[NetDev]
Name=$BR_IF
Kind=bridge
EOF

cat << EOF | sudo tee /etc/systemd/network/25-bridge-br0-zt.network
[Match]
Name=$ZT_IF

[Network]
Bridge=$BR_IF
EOF

cat << EOF | sudo tee /etc/systemd/network/25-bridge-br0-en.network
[Match]
Name=eth0 # might be en*

[Network]
Bridge=$BR_IF
EOF
```

Review configuration

```
tail -n+0 /etc/systemd/network/*
```

If needed, edit the files with the editor of your preference.

If it looks good:

```
sudo reboot
```

You should be able to, from the physical LAN, connect to the Pi via `$BR_ADDR`

If it takes a long time waiting for the network during boot

Sometimes the physical interface turns out to be a long "predictable interface name" like: "enb627eb0d4176", sometimes it's just `eth0`, depending on Raspbian version(???).

https://wiki.debian.org/NetworkConfiguration#Network_Interface_Names

Hook up a keyboard and monitor and check with ip addr then edit `/etc/systemd/network/25-bridge-br0-en.network` to match.

Configure ZeroTier Network

At `my.zerotier.com/network/$NETWORK_ID` -> `Settings` -> `Advanced`

- Delete the default Managed Route. Add the new Managed Route `$ZT_ROUTE`
- Change IPV4 Auto-Assign to Advanced
- Remove existing Pool. Create new Pool with start and end from `$ZT_POOL`
- For documentation purposes, assign `$BR_ADDR` to the ZeroTier bridge member

It should be working now.

Next steps

Either it worked, and you can ssh back in to `$BR_ADDR` after a minute, or it didn't work and the Pi isn't on the network anymore and you need to use the keyboard and monitor to figure out what went wrong.

> TIP
> Make a backup of the sd card so you don't have to repeat these steps

Appendix

Configure bridge with DHCP

```
cat << EOF | sudo tee /etc/systemd/network/25-bridge-br0.network
[Match]
Name=$BR_IF

[Network]
DHCP=yes
EOF
```

FAQ

I can ping the bridge, but not behind it

Sometimes, iptables rules apply: `echo "0" > /proc/sys/net/bridge/bridge-nf-call-iptables` or `iptables -A FORWARD -p all -i br0 -j ACCEPT`

See: https://serverfault.com/questions/162366/iptables-bridge-and-forward-chain

Why is the Managed Route /23 and the LAN subnet /24?

Say you have a laptop that is on the ZeroTier network and you bring it home. Now its WiFi address and ZeroTier address are in the same subnet. Which interface/address should your laptop use for internet access? https://en.wikipedia.org/wiki/Longest_prefix_match

Why is an app on my phone not working over ZeroTier?

Unfortunately the iOS and Android VPN APIs won't let ZeroTier use multicast/broadcast. These are typically how apps auto-discover services on the LAN.

References

- https://systemd.network/systemd.network.html
- https://hackaday.io/project/162164/instructions

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